

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA5607

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.* (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)

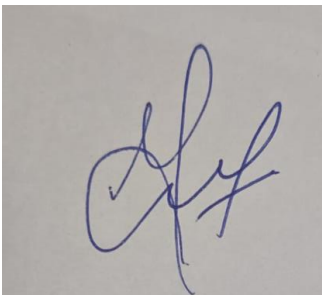
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

*I **M.Mukesh Balaji (192412156)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



1. Understanding of Design

The proposed model addresses the fronthaul link that connects the Remote Radio Head (RRH) at the antenna site to the Baseband Unit (BBU) or Distributed Unit (DU) in a centralized/Cloud RAN deployment. In legacy fronthaul implementations, the Common Public Radio Interface (CPRI) transports fully digitized, raw In-phase/Quadrature (IQ) radio samples over dedicated optical fiber, with no processing performed at the RRH itself. The design objective is to build an optimized fronthaul model that reduces this raw bit-rate burden while preserving the centralized processing benefits of C-RAN, particularly under 5G massive MIMO deployments where a large number of antenna elements multiplies the digitized sample stream. Key constraints include strict fronthaul latency budgets (well under 100 microseconds one-way for timing-critical HARQ operations), tight time and frequency synchronization between RRH and DU, and the need to keep the solution deployable over existing fiber and packet-switched infrastructure rather than requiring proprietary point-to-point links for every added antenna.

2. Evaluation of Design

Evaluating the traditional CPRI-based fronthaul reveals a clear structural weakness once massive MIMO is introduced. Its main strength is fidelity: because every raw IQ sample is transported unmodified, the DU retains full flexibility to run advanced baseband algorithms such as Coordinated Multipoint (CoMP) and joint interference cancellation across cells. However, CPRI's bit rate scales linearly with the number of antenna elements and channel bandwidth, independent of actual user traffic. For a 64-antenna massive MIMO array operating on a 100 MHz channel, this produces a fronthaul demand of several hundred Gbps, far beyond what a single high-capacity optical link can carry. This creates a severe design gap: the fronthaul, not the radio channel, becomes the network bottleneck, and scaling to more antennas or wider bandwidths quickly becomes economically and physically impractical over dedicated CPRI fiber.

3. Justification

The historical choice of raw-sample CPRI transport is justified for earlier-generation C-RAN because it kept the RRH extremely simple and inexpensive (essentially just an RF front end and digitizer), while concentrating all baseband intelligence, and therefore all vendor upgrades and interference-coordination gains, at the centralized DU/BBU pool. This was a sound trade-off when channel bandwidths and antenna counts were modest. In 5G, however, the same reasoning that justified centralization no longer justifies keeping the interface fully raw: the sample rate growth from massive MIMO and wide carrier bandwidths outpaces realistic fiber and switching capacity, so an optimized split that keeps centralization benefits while trimming the transported data becomes the stronger engineering choice.

4. Suggestions for Improvement

It is recommended to move from raw CPRI (equivalent to the lower PHY split, Option 8) to an eCPRI-based fronthaul following the 3GPP/O-RAN 7.2x functional split. In this split, the RRH performs the initial FFT, cyclic-prefix removal, and part of precoding/beamforming locally, so only frequency-domain, compressed IQ data for the active resource blocks is sent over the fronthaul instead of full time-domain raw samples. Combining this split with block-floating-point compression typically reduces the transported data rate by roughly an order of magnitude. In addition, migrating from dedicated point-to-point CPRI links to a packet-based eCPRI-over-Ethernet fronthaul allows statistical multiplexing across cells, use of standard 25G/100G switches instead of proprietary hardware, and easier scaling as antenna counts grow. Precision Time Protocol (PTP) with hardware timestamping should be retained to preserve the tight synchronization the split still requires.

5. Communication

This report is organized to move logically from the design context and constraints, through a critical evaluation of the baseline CPRI approach, to a justified and quantified improvement proposal. The accompanying simulation translates the qualitative argument into concrete bandwidth and latency figures, giving a clear, evidence-based basis for adopting the optimized eCPRI fronthaul model.

6. MATLAB Simulation Code

The following MATLAB code evaluates the fronthaul bandwidth and latency for a 64-antenna massive MIMO deployment, comparing the traditional raw-CPRI model against the proposed optimized eCPRI (7.2 split) model.

```
% MATLAB Simulation: Traditional CPRI vs Optimized eCPRI Fronthaul
% Parameters
num_antennas = 64;           % Massive MIMO array elements at gNB
cpri_rate_per_antenna = 9.8; % Raw digitized IQ rate per antenna (Gbps, 100MHz)
link_capacity = 100;         % Standard fronthaul aggregation link (100GE, Gbps)

% Latency parameters (in microseconds)
t_prop_cpri = 5; % Fiber propagation delay, RRH to BBU
t_proc_cpri = 2; % Digitization/framing delay (no local processing)
t_prop_ecpri = 5; % Fiber propagation delay, RRH to DU
t_proc_ecpri = 8; % Local FFT/precoding delay at RRH (Option 7.2 split)

disp('--- Fronthaul Link Simulation ---');

% 1. Traditional CPRI (Option 8 - Full Raw IQ)
total_cpri_rate = num_antennas * cpri_rate_per_antenna;
fprintf('--- Traditional CPRI Model ---\n');
fprintf('Total Fronthaul Data Rate: %.1f Gbps\n', total_cpri_rate);
if total_cpri_rate > link_capacity
    fprintf('STATUS: LINK OVERLOAD (Exceeded by %.1f Gbps)\n', total_cpri_rate -
link_capacity);
    fprintf('Parallel 100GE Links Required: %.1f\n', total_cpri_rate/link_capacity);
```

```

else
    fprintf('STATUS: Bandwidth Adequate\n');
end
fprintf('End-to-End Fronthaul Latency: %d us\n\n', t_prop_cpri + t_proc_cpri);

% 2. Optimized eCPRI Model (Implementation of Suggestion)
% Assumption: 7.2 functional split + frequency-domain compression
compression_factor = 10;
ecpri_rate_per_antenna = cpri_rate_per_antenna / compression_factor;
total_ecpri_rate = num_antennas * ecpri_rate_per_antenna;

fprintf('--- Optimized eCPRI Model (7.2 Split) ---\n');
fprintf('Total Fronthaul Data Rate: %.2f Gbps\n', total_ecpri_rate);
if total_ecpri_rate > link_capacity
    fprintf('STATUS: LINK OVERLOAD\n');
else
    fprintf('STATUS: Bandwidth Adequate (fits single 100GE link)\n');
end
fprintf('End-to-End Fronthaul Latency: %d us\n', t_prop_ecpri + t_proc_ecpri);
fprintf('Bandwidth Reduction: %.1f%%\n', (1 - total_ecpri_rate/total_cpri_rate)*100);

```

OUTPUT:

```

>> fronthaul_sim
--- Fronthaul Link Simulation ---
--- Traditional CPRI Model ---
Total Fronthaul Data Rate: 627.2 Gbps
STATUS: LINK OVERLOAD (Exceeded by 527.2 Gbps)
Parallel 100GE Links Required: 6.3
End-to-End Fronthaul Latency: 7 us

--- Optimized eCPRI Model (7.2 Split) ---
Total Fronthaul Data Rate: 62.72 Gbps
STATUS: Bandwidth Adequate (fits single 100GE link)
End-to-End Fronthaul Latency: 13 us
Bandwidth Reduction: 90.0%
>>

```

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning,	Provides justification with some	Weak or unclear justification	No justification for feedback
		evidence, and relevant principles	supporting evidence		
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation